CONSTRUCTIBLY CLOSED SETS STABLE UNDER SPECIALIZATION ARE CLOSED, AND DUALLY FOR OPEN

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The statement that the constructible topology of a spectral space is Hausdorff and quasi-compact. Invoked inside the counted proof (source lines 4698-4700). Makes the total intersection of the neighbourhood pieces nonempty.
- The statement that an intersection of closed subsets of a quasi-compact space is nonempty when all finite intersections are. Invoked inside the counted proof (source line 4700). Produces a point of the intersection.
- The statement that the complement of a constructible set is constructible and closed under the dual stability. Invoked inside the counted proof (source lines 4708-4710). Reduces the open case to the closed one.

Source reference. Original lemma lemma-constructible: sources/stacks-topology/pinned-original-source.tex, source lines 2788–2788. Navigation pointer only: the counted text refers to the Stacks item lemma-constructible, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original lemma lemma-constructible-hausdorff-quasi-compact: sources/stacks-topology/pinned-original-source.tex, source lines 4522–4522. Navigation pointer only: the counted text refers to the Stacks item lemma-constructible-hausdorff-quasi-compact, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original lemma lemma-intersection-closed-in-quasi-compact: sources/stacks-topology/pinned-original-source.tex, source lines 1941–1941. Navigation pointer

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Source reference. Original lemma lemma-open-closed-specialization: sources/stacks-topology/pinned-original-source.tex, source lines 3785–3785. Navigation pointer only: the counted text refers to the Stacks item lemma-open-closed-specialization, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Lemma 0.1. *Let X be a spectral space. Let $E \subset X$ be a subset closed in the constructible topology (for example constructible).*

(1) *If $x \in \overline{E}$, then x is the specialization of a point of E.*
(2) *If E is stable under specialization, then E is closed.*
(3) *If $E' \subset X$ is open in the constructible topology (for example constructible) and stable under generalization, then E' is open.*

Proof. Proof of (1). Let $x \in \overline{E}$. Let $\{U_i\}$ be the set of quasi-compact open neighbourhoods of x. A finite intersection of the U_i is another one. The intersection $U_i \cap E$ is nonempty for all i. Since the subsets $U_i \cap E$ are closed in the constructible topology we see that $\bigcap (U_i \cap E)$ is nonempty by Lemma lemma-constructible-hausdorff-quasi-compact and Lemma lemma-intersection-closed-in-quasi-compact. Since $\{U_i\}$ is a fundamental system of open neighbourhoods of x, we see that $\bigcap U_i$ is the set of generalizations of x. Thus x is a specialization of a point of E.

Part (2) is immediate from (1).

Proof of (3). Assume E' is as in (3). The complement of E' is closed in the constructible topology (Lemma lemma-constructible) and closed under specialization (Lemma lemma-open-closed-specialization). Hence the complement is closed by (2), i.e., E' is open. □